



Cross-Document Validation using Document Intelligence

Milestone 2 : Data Collection and Data Preparation

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Milestone Objective

Prepare datasets required to evaluate a system that can perform :

- Document entity extraction
- Cross-document validation
- Semantic equivalence detection
- Numeric normalization
- Conflict detection

Two datasets were prepared to support these tasks:

1. FUNSD Dataset
2. Healthcare SBC & Benefit Grid Dataset

An illustration of a white document with a blue magnifying glass over the date field. The document has a blue header bar, a blue box containing 'Name: John Doe', and an orange box containing 'Date: 12/05/2021'. The magnifying glass is blue and is positioned over the date field. The background is a blue gradient with a white circular shape on the left side.

Document Entity Extraction

Dataset Overview

Two complementary datasets were used:

1. FUNSD Dataset

- Form understanding
- Key-value extraction
- Document entity recognition

2. SBC & Benefit Grid Dataset

- Cross-document validation
- Coverage comparison
- Cost-sharing verification

Together they simulate real-world document validation workflows.

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FILTER SCORES

Brand: RALEIGH (BELAIR portion not tested) Project #: 1969-105
Commercial: LAKE - NEW PACK :40 (with BELAIR Badminton :20) Sample: 336
Code #: BW-RT-69-98 PM6 Base: (234)
Supplier: AUDIENCE STUDIES

TEST DATES

L. Angeles: 8/5 and 6
Chicago: 8/8

PM6
SCORES

COMMENTS

Overall 1.7

CITY
Los Angeles 0.0
Chicago 3.3

SEX
Male 0.0
Female 3.3

AGE
16-25 0.0
26-35 0.0
36-45 0.0
46 & Over 9.3

35 & Under 0.0
36 & Over 5.0

This commercial was tested in color.

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FUNSD Dataset

FUNSD (Form Understanding in Noisy Scanned Documents)

Key characteristics:

- Scanned form documents
- Layout-aware annotations
- Key-value entity pairs
- Real-world document noise

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FUNSD DATA PREPARATION

Steps performed:

1. Selected sample documents from FUNSD dataset
2. Identified logical document sections
3. Defined entity groups within each section

Example entities:

- Name
- Address
- Date
- Numeric values
- Identifiers

Prepared dataset for entity extraction experiments.

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FUNSD Data Augmentation

To simulate real-world document variations:

- Several modifications were introduced:
- Synonym replacement
- Numeric format variations
- Layout modifications
- OCR-style noise

Examples:

- 11% → 0.11
- Interest Rate → Applicable Interest

Purpose: Test robustness of entity extraction models.

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SBC Dataset

Summary of Benefits and Coverage (SBC) document will help you choose a health [plan](#). The SBC shows you how you and the [plan](#) would pay for covered health care services. NOTE: Information about the cost of this [plan](#) (called the [premium](#)) will be provided separately. For a summary. For more information about your coverage, or to get a copy of the complete terms of coverage, [insert contact information]. For common terms, such as [allowed amount](#), [balance billing](#), [coinsurance](#), [copayment](#), [deductible](#), [provider](#), or other underlined terms see the Glossary at [www.\[insert\].com](#) or call 1-800-[insert] to request a copy.

Summary of Benefits and Coverage (SBC) documents

Healthcare insurance documents containing:

- Coverage information
- Cost-sharing details
- Healthcare service descriptions

Typical document size: 2 to 11 pages

These documents serve as Doc A in the dataset.

Q	Answers	Why This Matters:
What is the deductible?	\$500/Individual or \$1,000/family	Generally, you must pay all of the costs from providers up to the deductible amount before this plan begins to pay. If you have other family members on the plan , each family member must meet their own individual deductible until the total amount of deductible expenses paid by all family members meets the overall family deductible .
What services are covered before you meet your deductible?	Yes. Preventive care and primary care services are covered before you meet your deductible .	This plan covers some items and services even if you haven't yet met the deductible amount. But a copayment or coinsurance may apply. For example, this plan covers certain preventive services without cost sharing and before you meet your deductible . See a list of covered preventive services at https://www.healthcare.gov/coverage/preventive-care-benefits/ .
What are the specific deductibles for prescription drug coverage and occupational therapy services?	Yes. \$300 for prescription drug coverage and \$300 for occupational therapy services.	You must pay all of the costs for these services up to the specific deductible amount before this plan begins to pay for these services.
What is the out-of-pocket limit?	For network providers \$2,500 individual / \$5,000 family; for out-of-network providers \$4,000 individual / \$8,000 family	The out-of-pocket limit is the most you could pay in a year for covered services. If you have other family members in this plan , they have to meet their own out-of-pocket limits until the overall family out-of-pocket limit has been met.
What is the out-of-pocket limit for copayments, premiums, balance-billing charges, and health care this plan doesn't cover?	Copayments for certain services, premiums , balance-billing charges, and health care this plan doesn't cover.	Even though you pay these expenses, they don't count toward the out-of-pocket limit .
What is the provider network?	Yes. See www.[insert].com or call 1-800-[insert] for a list of network providers .	This plan uses a provider network . You will pay less if you use a provider in the plan's network . You will pay the most if you use an out-of-network provider , and you might receive a bill from a provider for the difference between the provider's charge and what your plan pays (balance billing). Be aware, your network provider might use an out-of-network provider for some services (such as lab work). Check with your provider before you get services.
Do I need a referral to see a specialist?	Yes.	This plan will pay some or all of the costs to see a specialist for covered services but only if you have a referral before you see the specialist .

SBC Data Collection

SBC documents were collected by searching for:

- Highmark PPO insurance plans

Challenges encountered:

- Duplicate documents
- Different formatting styles

Solutions:

Searching across plan tiers

- Gold
- Silver
- Bronze

Using multiple plan identifiers

Total collected: 20 SBC documents



Benefit Grid Creation

To standardize SBC information: A Benefit Grid template was created.

The grid includes columns for:

- Service Name
- In-Network Cost
- Out-of-Network Cost
- Notes

Example services:

- Doctor visits
- Emergency care
- Hospital stay
- Prescription drugs

Each SBC was converted into a Benefit Grid document (Doc B).

Document Pair Structure

Each dataset sample consists of:

Doc A: SBC Document

Doc B: Benefit Grid

Example relationship:

SBC Document → Detailed explanation

Benefit Grid → Structured summary

This structure enables cross-document validation.

Data Augmentation (Benefit Grid)

Controlled modifications were applied to simulate real-world inconsistencies.

Out of 20 benefit grids:

- 12 kept unchanged
- 8 intentionally modified

Modifications were applied using PDF editing tools.

Augmentation Types

Semantic Changes

Example: "No charge" → "Covered in full"

Numeric Format Changes

Example: "\$6,550" → "6550"

Value Conflicts

Example: "\$525 copay" → "\$400 copay"

Coverage Changes

Example: "\$30 copay" → "Fully covered"

Ground Truth JSON Creation

Ground truth JSON files were created for augmented document pairs.

Each JSON contains:

- Entity name
- Value in Doc A
- Value in Doc B
- Normalized value
- Validation type
- Match or mismatch result

Also includes perturbation logs to track modifications.

Validation Scenarios Covered

Dataset supports multiple validation cases:

1. Exact Match
2. Semantic Equivalence
3. Numeric Normalization
4. OCR Noise Handling
5. Conflict Detection
6. Coverage Change Detection

These scenarios represent real-world document inconsistencies.

Dataset Output

Final dataset produced:

Dataset Component	Count
FUNSD Samples	Multiple
SBC Documents (Doc A)	20
Benefit Grid Documents (Doc B)	20
Ground Truth JSON Files	8

This dataset supports both:

- Entity extraction
- Cross-document validation

Challenges Faced

During dataset preparation:

- Duplicate Documents Many identical SBC files online.
- Complex Tables Multiple network tiers in SBC tables.
- Formatting Issues PDF extraction caused misaligned columns.
- Ambiguous Service Descriptions Some rows combined multiple services.
- Manual verification was performed to resolve these issues.

Team Contributions

Karthik

Researched the validation use case, Identified FUNSD and SBC datasets, Designed augmentation scenarios

Mayank

Worked on FUNSD data augmentation, Created modified document samples

Ashish

Performed SBC and Benefit Grid augmentation

Generated ground truth JSON files

Mallesha

Documentation.

Milestone Summary

Milestone 2 successfully achieved:

- ✓ Dataset research and selection
- ✓ Data collection from multiple sources
- ✓ Document standardization
- ✓ Controlled data augmentation
- ✓ Ground truth generation

Prepared dataset enables:

- ✓ Entity extraction evaluation
- ✓ Cross-document validation experiments

Thank You :)